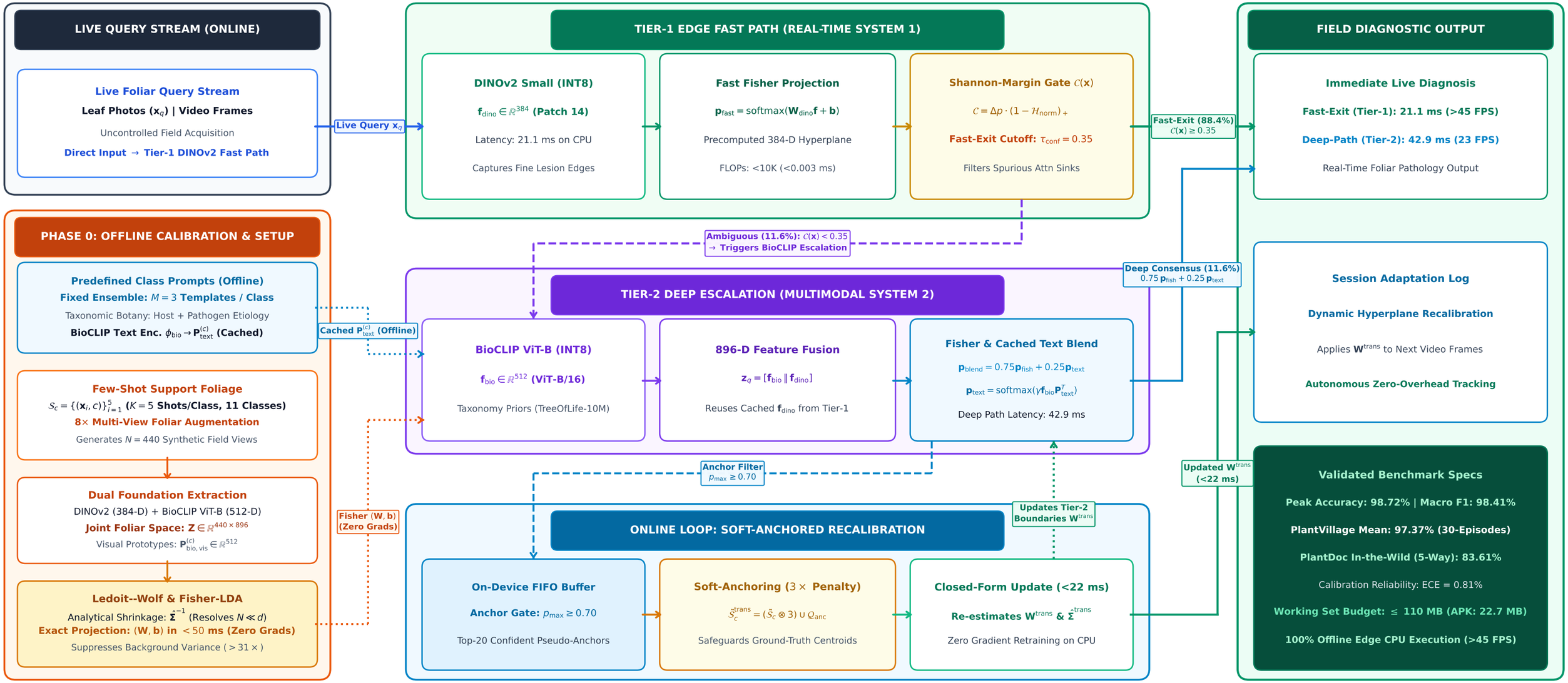

LIVE QUERY STREAM (ONLINE)
Live Foliar Query Stream
Leaf Photos ($\mathbf{x}_q$) | Video Frames
Uncontrolled Field Acquisition
Direct Input → Tier-1 DINOv2 Fast Path
Live Query $\mathbf{x}_q$
TIER-1 EDGE FAST PATH (REAL-TIME SYSTEM 1)
DINOv2 Small (INT8)
$\mathbf{f}_{\text{dino}} \in \mathbb{R}^{384}$ (Patch 14)
Latency: 21.1 ms on CPU
Captures Fine Lesion Edges
Fast Fisher Projection
$\mathbf{p}_{\text{fast}} = \text{softmax}(\mathbf{W}_{\text{dino}}\mathbf{f} + \mathbf{b})$
Precomputed 384-D Hyperplane
FLOPs: <10K (<0.003 ms)
Shannon-Margin Gate $\mathcal{C}(\mathbf{x})$
$\mathcal{C} = \Delta p \cdot (1 - \mathcal{H}_{\text{norm}})_+$
Fast-Exit Cutoff: $\tau_{\text{conf}} = 0.35$
Filters Spurious Attn Sinks
Fast-Exit (88.4%)
$\mathcal{C}(\mathbf{x}) \geq 0.35$
FIELD DIAGNOSTIC OUTPUT
Immediate Live Diagnosis
Fast-Exit (Tier-1): 21.1 ms (>45 FPS)
Deep-Path (Tier-2): 42.9 ms (23 FPS)
Real-Time Foliar Pathology Output
Ambiguous (11.6%): $\mathcal{C}(\mathbf{x}) < 0.35$
→ Triggers BioCLIP Escalation
PHASE 0: OFFLINE CALIBRATION & SETUP
Predefined Class Prompts (Offline)
Fixed Ensemble: $M = 3$ Templates / Class
Taxonomic Botany: Host + Pathogen Etiology
BioCLIP Text Enc. $\phi_{\text{bio}} \rightarrow \mathbf{P}_{\text{text}}^{(c)}$ (Cached)
Cached $\mathbf{P}_{\text{text}}^{(c)}$ (Offline)
TIER-2 DEEP ESCALATION (MULTIMODAL SYSTEM 2)
BioCLIP ViT-B (INT8)
$\mathbf{f}_{\text{bio}} \in \mathbb{R}^{512}$ (ViT-B/16)
Taxonomy Priors (TreeOfLife-10M)
896-D Feature Fusion
$\mathbf{z}_q = [\mathbf{f}_{\text{bio}} \| \mathbf{f}_{\text{dino}}]$
Reuses Cached $\mathbf{f}_{\text{dino}}$ from Tier-1
Fisher & Cached Text Blend
$\mathbf{p}_{\text{blend}} = 0.75\mathbf{p}_{\text{fish}} + 0.25\mathbf{p}_{\text{text}}$
$\mathbf{p}_{\text{text}} = \text{softmax}(\gamma \mathbf{f}_{\text{bio}} \mathbf{P}_{\text{text}}^T)$
Deep Path Latency: 42.9 ms
Deep Consensus (11.6%)
$0.75\,\mathbf{p}_{\text{fish}} + 0.25\,\mathbf{p}_{\text{text}}$
Few-Shot Support Foliage
$S_c = \{(\mathbf{x}_i, c)\}_{i=1}^{5}$ ($K = 5$ Shots/Class, 11 Classes)
8× Multi-View Foliar Augmentation
Generates $N = 440$ Synthetic Field Views
Dual Foundation Extraction
DINOv2 (384-D) + BioCLIP ViT-B (512-D)
Joint Foliar Space: $\mathbf{Z} \in \mathbb{R}^{440 \times 896}$
Visual Prototypes: $\mathbf{P}_{\text{bio, vis}}^{(c)} \in \mathbb{R}^{512}$
Ledoit--Wolf & Fisher-LDA
Analytical Shrinkage: $\hat{\mathbf{\Sigma}}^{-1}$ (Resolves $N \ll d$)
Exact Projection: $(\mathbf{W}, \mathbf{b})$ in < 50 ms (Zero Grads)
Suppresses Background Variance (> 31 ×)
Fisher $(\mathbf{W}, \mathbf{b})$ (Zero Grads)
Session Adaptation Log
Dynamic Hyperplane Recalibration
Applies $\mathbf{W}^{\text{trans}}$ to Next Video Frames
Autonomous Zero-Overhead Tracking
Anchor Filter
$p_{\max} \geq 0.70$
Updated $\mathbf{W}^{\text{trans}}$ (<22 ms)
ONLINE LOOP: SOFT-ANCHORED RECALIBRATION
Updates Tier-2 Boundaries $\mathbf{W}^{\text{trans}}$
On-Device FIFO Buffer
Anchor Gate: $p_{\max} \geq 0.70$
Top-20 Confident Pseudo-Anchors
Soft-Anchoring (3 × Penalty)
$\tilde{S}_c^{\text{trans}} = (\tilde{S}_c \otimes 3) \cup \mathcal{Q}_{\text{anc}}$
Safeguards Ground-Truth Centroids
Closed-Form Update (<22 ms)
Re-estimates $\mathbf{W}^{\text{trans}}$ & $\tilde{\mathbf{\Sigma}}^{\text{trans}}$
Zero Gradient Retraining on CPU
Validated Benchmark Specs
Peak Accuracy: 98.72% | Macro F1: 98.41%
PlantVillage Mean: 97.37% (30-Episodes)
PlantDoc In-the-Wild (5-Way): 83.61%
Calibration Reliability: ECE = 0.81%
Working Set Budget: ≤ 110 MB (APK: 22.7 MB)
100% Offline Edge CPU Execution (>45 FPS)